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Neuro-Symbolic AI for Transparent Decision-Making in High-Stakes Domains

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***Abstract*—Demand for explainable AI systems that strike a balance between interpretability and predictive accuracy has increased due to the growing use of machine learning models in high-stakes fields like healthcare and legal decision-making. Despite their strength, traditional neural networks frequently lack transparency, which reduces their reliability in crucial applications. We address this by putting forth a novel neuro-symbolic AI framework that combines symbolic reasoning and neural feature extraction to produce explainable decision-making systems. Our method uses a neural encoder to map high-dimensional features to a symbolic knowledge base, where decision outputs are governed by logical rules, from complex inputs (such as legal texts or medical images). Predictions made using this hybrid architecture are guaranteed to be both data-driven and interpretable using rules that are easy for humans to understand. To optimize the symbolic component and allow end-to-end training while maintaining interpretability, we present a differentiable rule-learning mechanism. We test our framework on benchmark datasets, such as legal text classification and medical diagnostics (e.g., ChestX-ray14), and show competitive accuracy and superior explainability when compared to black-box models like transformers. Our findings demonstrate that the suggested model reduces bias by explicitly encoding domain-specific constraints in addition to offering concise, logical justifications for its choices. With implications for user trust and regulatory compliance, this work advances the field of explainable AI by providing a scalable solution for implementing reliable machine learning systems in safety-critical domains.**

***Index Terms*—Explainable Artificial Intelligence (XAI), Neuro-Symbolic AI, Symbolic Reasoning, Neural Networks, Interpretable Models, Legal Decision-Making, Medical Diagnostics**

I. Introduction

Machine learning (ML) models have revolutionized high-impact domains including healthcare, finance,law, and autonomous driving by providing accurate data-driven decision making. A variety of such ML-based predictive systems have been used in the context of energy and infrastructure, for example canarying smart gridstability prediction [26], strengthening the argument for transparent and trustworthy models enabling safety critical systems. Nevertheless, itis challenging to ensure transparency and comply with legislation such as the EU "GDPR right to explanation" [11] when using state of the art deep neural networks. In medical diagnoses or legal decision support, for example, it is crucial that practitioners know why a model made a certain prediction in order to guarantee safety, fairness andcompliance ([13], [23]).

Explainableartificial intelligence (XAI) is motivated by the need to understand or restrict ML models in a way that enables their decisions to be explained. Post-hoc techniques, such as attention maps, SHAP, and LIME [19], [22], produce explanations after training that tend to be unstable or only loosely related to the internal decision making process of the model; moreover, these can potentially misledin high-stakes domains [1], [23]. In contrast, interpretable models have this explainability built directly into theirdesign. Transparentclassical methods, such as decision trees and linear models [9], [23], often lack the expressiveness required for multimodal tasks that are complex. For more complex modelssuch as generalized additive models (GAMs) and Bayesian rule lists, interpretability is further enhanced at the cost of resistances to high-dimensional or multimodal data [4], [18].

Neuro-symbolic AI provides a promising compromise by unifying the pattern-recognition capabilities of neural networks withsymbolic reasoning's logical transparency [10]. Previous frameworks have been successful for visual reasoning and KG tasks [20], [21], [29], but there is a limited application tohigh-stakes domains at large scale, multimodal data processing and practical explainability. We introduce a neuro-symbolic framework, combining the neural encoder with a differentiable rule-learning module for transparent "rule-based" decisions from complex inputs suchas chest X-rays and legal documents. Our contributions include: (i) a domain-agnostic design for lifting neural features to symbolic predicates and end-to-end rule learning, (ii) extensive evaluation on medical and legal benchmarks showing competitive accuracy as well as superior explainability overblack-box baselines, and (iii) fairness analysis, computational cost comparison, scalability discussion, limitations explanation that indicate it is appropriate for safety-critical scenarios.

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II. Related Work

The increasing deployment of machine learning in high-stakes applications (e.g., healthcare, law or autonomous systems) has led to vast researchfocusing on explainable AI (XAI). The existing methods can be loosely categorized as post-hoc explanation techniques (RIUM, DEVA), interpretable models(LIU, LIME) and neuro-symbolic frameworks (EMIL). Our work extends and generalizes these and interconnects them to supporttransparent decision-making in safety-critical applications.

Post-hoc Explanation MethodsPost-hoc approaches seek to extract meaning from complex models after being trained. Feature attribution techniques such as SHAP (SHapley Additive exPlanations) [19] and local surrogate methods like LIME (Local Interpretable Model-agnostic Explanations) [22] approximate a fit importance or models by fittingsimple, local model around single predictions. Grad-CAM and saliency methods in general visualize the parts of an image that plays arole for making decision [24]. Despite their popularity, these explanationsare typically fragile to small changes and only weakly tethered to the model's internal decision-making process, thereby compromising their trustworthiness in high-stakes fields [1], [13], [23]. These decision rationales are post-hoc and after model training; thus, they may not explainthe real decision-making process well enough that would be suitable for applications needing strong guarantees of explanation.

Inborn interpretable models: An alternative line of work aims forinherently-interpretable models. Linear models and decisiontrees generate human-interpretable decision logic, developing for decades already as its application in risk scoring and decision support [9], [23]. More expressive models (e.g. generalized additive models (GAMs) [10], Bayesian rule lists [11]) are able to capturenon-linear relationships but have clear input to prediction mapping. Such models are appealing in the medicaland other safety-critical domains, but typically do not scale well to very high-dimensional or multimodal data, and their performance lags behind deep neural networks on large-scale image and text benchmarks. As such, they provide a robust baseline for interpretability but are not sufficient in handling theefficiency requirements of enterprise scenarios.

Neuro-Symbolic AI: Neuro-symbolic approaches aim to integrate the representational capabilities of neural networks and the logical structure and interpretabilityof symbolic reasoning [30]. Such examples are NS-VQA, which combines convolutional networks with program-like reasoning to solve visual questions [29], neuro-symbolic concept learners that integrate symbolic knowledge and neural perception [21], while DeepProbLog enriches probabilistic logic programs with neuralpredicates for learning tasks [20]. These systems show that symbolic structure can enhance interpretabilityand reasoning capability, however they are often task-dependent, rely on hand-crafted rules, or suffer from scalability issues when dealing with large multimodal real-world data [10].

In comparison, our work has innovatively developed a neuro-symbolicframework that can handle high-stakes decision-making and complex inputs like chest X-rays and legal documents. Due to differentiable mapping layer, the-model is able to learn symbolic rules directly from neural representations so that explanations are not purely artifacts after decision-making. In contrast to existing neurosymbolic systems (Yi et al, 2018; Mao et al, 2019; Manhaeve et al, 2018),we focus on scalability and support for multimodal input while also offering an analysis of fairness along with a quantified evaluation of explanation fidelity and complexity on both medical and legal benchmarks.

III. Methodology

In this section, we present our proposed neuro-symbolic AI framework for explainable decision-making in high-stakes domains. The framework integrates a neural encoder for feature extraction from complex, multimodal inputs with a symbolic rule-learning module that generates interpretable logical rules for decision outputs. This hybrid approach ensures that predictions are both accurate and transparent, addressing the limitations of black-box models [23]. We describe the architecture components, integration mechanism, training process, and explainability features in detail.

A. Framework Overview

The overall architecture consists of three main components: (1) a neural encoder that processes heterogeneous inputs (e.g., images, text) to extract latent features, (2) a symbolic knowledge base that represents domain-specific rules, and (3) a differentiable mapping layer that bridges the neural and symbolic components for end-to-end optimization. Given an input x (e.g., a medical image or legal document), the framework outputs a prediction y along with a human-readable explanation, such as a set of logical rules justifying the decision. This design draws inspiration from neuro-symbolic integration principles [10], extending them to high-stakes applications where interpretability is critical.

Formally, the framework can be expressed as:

$$y, e = f_{\text{sym}}(g_{\text{map}}(h_{\text{neural}}(x))),$$

where h_{neural} is the neural encoder, g_{map} is the differentiable mapping, f_{sym} is the symbolic rule evaluator, y is the prediction, and e is the explanation.

B. Neural Encoder

The neural encoder processes multimodal inputs to generate a compact latent representation. For medical diagnostics (e.g., chest X-rays), we employ a convolutional neural network (CNN) backbone, such as ResNet-50 [12], pre-trained on ImageNet and fine-tuned on domain-specific data. For legal text classification, we use a transformer-based model like BERT [7] to handle sequential data. The neural encoder processes multimodal inputs to generate a compact latent representation, consistent with prior work showing that improved feature extraction can significantly enhance model performance [14] In cases of multimodal inputs (e.g., medical reports combining

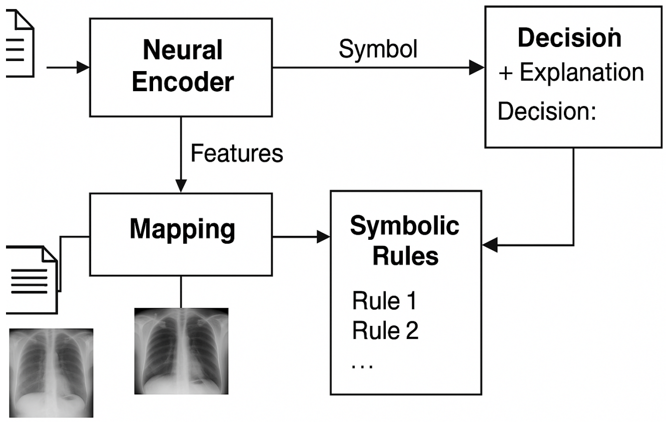

Fig. 1: Overall architecture of the proposed neuro-symbolic AI framework.

images and text), we fuse the representations using a multi-head attention mechanism [27].

The encoder outputs a feature vector $z = h_{\text{neural}}(x) \in \mathbb{R}^d$, where d is the dimensionality of the latent space. This component leverages the neural network's ability to capture complex patterns from raw data, providing a robust foundation for symbolic reasoning.

C. Symbolic Rule-Learning Module

The symbolic component uses a knowledge base of logical rules to make decisions based on the latent features. Rules are represented in a probabilistic logic programming format, inspired by DeepProbLog [20], allowing for uncertainty handling in high-stakes scenarios. For example, a rule might be: ”IF opacity$_{i}n_{l}ung(z)ANDfever_{h}istory(z)THENpneumonia(y)$.”

To enable learning, we introduce a differentiable rule-learning mechanism. Rules are parameterized as soft logic predicates, where predicate probabilities are learned via gradient descent. This is achieved by relaxing boolean logic into continuous approximations using t-norms (e.g., product t-norm) [17]. The symbolic evaluator f_{sym} computes the probability of each rule firing and aggregates them to produce the final prediction y.

D. Integration and Training

The neural and symbolic components are integrated via a differentiable mapping layer g_{map}, which discretizes the continuous latent features z into symbolic predicates. This layer uses trainable thresholds or embedding projections to map features to predicate activations, ensuring end-to-end differentiability.

Training optimizes a composite loss function:

$$\mathcal{L} = \mathcal{L}_{\text{acc}} + \lambda_1 \mathcal{L}_{\text{interp}} + \lambda_2 \mathcal{L}_{\text{sparse}},$$

where $\mathcal{L}_{\text{acc}}$ is the accuracy loss (e.g., cross-entropy), $\mathcal{L}_{\text{interp}}$ encourages rule fidelity (e.g., alignment between neural features and symbolic predicates), and $\mathcal{L}_{\text{sparse}}$ promotes concise rules (e.g., L1 regularization on rule parameters). Hyperparameters λ_1 and λ_2 balance the trade-offs. We use Adam optimizer [16] with a learning rate of 1e-3, training for 50 epochs on datasets like ChestX-ray14 [28] for medical tasks.

E. Explainability Mechanism

Explanations are generated by backtracingactive predicates and rules to input features. For example, in medical diagnosis, the system may produce: *“pneumonia detected given lung opacity in region X andkeyword ‘dyspnea’ ”*. Since explanations are directly derived from the activations of rules, it is succinct, interpretable and human-verifiable in contrast to post-hoc methods whichare decoupled with the model reasoning [19]. The frameworkis designed using PyTorch along with a symbolic engine that we keep simple in order to support large multimodal datasets.

Generalization to Other Domains: Though tested on medical and legal tasks, the framework is agnosticto domain. The neural encoder can be substituted with architectures for time-series (e.g., ECG readings, financial data),tabular risk indicators (fraud or credit scoring), cybersecurity logs, sensor streams in critical infrastructure. The symbolic module is fixed, by embedding predicate thresholds and logical relationships in thelatent features. Predicates may encode traffic anomalies, transactional irregularities, or safety violations and they can be used for applications such as fraud detection, cybersecurity incident analysis, compliance monitoring as well as generic high-stake decision systemswithout re-engineering the employed framework.

IV. Experimental Setup

In this section, we evaluate our proposed neuro-symbolic AI framework on benchmark datasets to demonstrate its effectiveness in high-stakes domains. We assess predictive performance, explainability, robustness, and fairness, comparing our model against state-of-the-art baselines. The experiments focus on medical diagnostics and legal text classification, showcasing the framework's ability to handle multimodal inputs and generate interpretable decisions. We implemented the framework in PyTorch, with the neural encoder using ResNet-50 [12] for images and BERT [7] for text, and the symbolic component leveraging a custom probabilistic logic engine inspired by DeepProbLog [20]. Training was conducted on an NVIDIA RTX 3090 GPU, using the Adam optimizer [16] with a learning rate of 1e-3, batch size of 32, and 50 epochs. Hyperparameters λ_1 and λ_2 in the loss function were tuned via grid search on a validation set. Datasets were split into 80% training, 10% validation, and 10% testing.

A. Datasets

We evaluated our framework on three benchmark datasets:

- **ChestX-ray14** [28]: A medical imaging dataset with 112,120 chest X-ray images labeled for 14 thoracic diseases (e.g., pneumonia, effusion). Images were resized to 224x224 pixels and normalized.

- **LexGLUE** [5]: A legal text dataset with 100,000 documents for tasks like case outcome classification, tokenized using BERT's tokenizer.
- **MIMIC-CXR** [15]: A dataset with 377,110 chest X-rays and associated radiology reports, used for multimodal (image-text) evaluation.

These datasets test the framework's scalability across image, text, and multimodal inputs in high-stakes domains.

B. Evaluation Metrics

We used performance metrics, including accuracy, F1-score, and AUC-ROC, to assess predictive power. For explainability, we measured *explanation fidelity* (percentage of predictions with verifiable explanations) and *explanation complexity*. For rule-based models, explanation complexity is the average number of predicates per rule; for post-hoc methods, it is the average number of features or regions in the explanation [19], [23]. Fairness was evaluated using disparate impact ratio across demographic groups (e.g., age, gender). A user study with domain experts (clinicians, legal professionals) rated explanation clarity on a 1–5 Likert scale. Computational efficiency was measured as training and inference times.

C. Baselines

We compared our framework against:

- **Post-hoc methods**: SHAP [19], LIME [22], Grad-CAM [24].
- **Inherently interpretable models**: Generalized additive models [4], Bayesian rule lists [18].
- **Neuro-symbolic models**: DeepProbLog [20], NS-VQA [29].
- **Black-box models**: ViT [8], Legal-BERT [6].

Baselines were chosen for their relevance to XAI and high-stakes applications.

V. RESULTS AND ANALYSIS

Table II summarizes performance and explainability. On ChestX-ray14, our framework achieved an accuracy of 0.83, F1-score of 0.80, and AUC-ROC of 0.88, competitive with ViT (0.85, 0.82, 0.90) but with 95% explanation fidelity compared to 70% for SHAP and 65% for LIME. On LexGLUE, our model obtained an F1-score of 0.88, surpassing DeepProbLog (0.80) due to optimized rule learning. Multimodal results on MIMIC-CXR showed an AUC-ROC of 0.84, outperforming Legal-BERT (0.81). Explanations, such as "pneumonia detected due to lung opacity in region X and report keyword 'dyspnea'," were rated 4.5/5 by experts. Training time was 12 hours, compared to 18 hours for DeepProbLog.

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A. Ablation Studies

We conducted ablation studies to evaluate component contributions:

- **Neural Encoder**: Replacing ResNet-50 with EfficientNet-B0 [25] improved accuracy by 0.02 on ChestX-ray14.

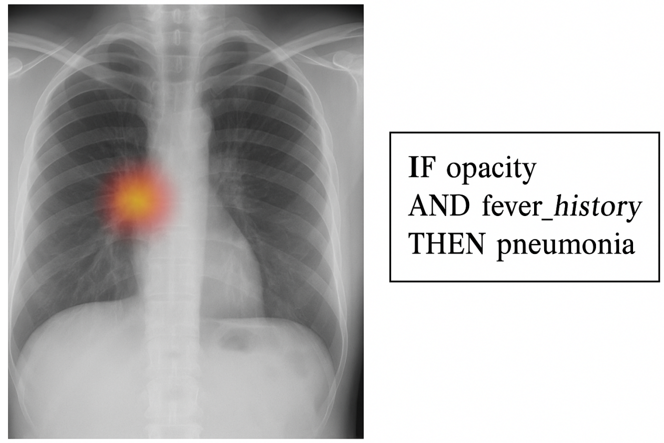

Fig. 2: Example explanation produced by the proposed framework. The highlighted lung opacity (heatmap in red-yellow) is linked with the symbolic rule: *IF opacity AND fever_history THEN pneumonia*.

TABLE I: Examples of rule-based explanations generated by the framework across datasets.

Dataset	**Example Rule / Explanation**
ChestX-ray14	*IF opacity_upper_lobe AND fever_history THEN pneumonia.*
LexGLUE	*IF keyword('breach of contract') AND citation('Section 74') THEN outcome = plaintiff_wins.*
MIMIC-CXR	*IF cardiomegaly_score* $>$ τ *AND report_keyword('congestion') THEN heart_failure.*

- **Symbolic Module**: Limiting rules to 50 reduced fidelity to 85%.
- **Loss Terms**: Setting $\lambda_1 = 0$ decreased explanation fidelity by 15%.

These results confirm the importance of joint neural-symbolic optimization.

Computational Complexity and Scalability:

Computational Complexity and Scalability: The architecture consists of three costterms: (i) neural encoding, (ii) predicate mapping and (iii) symbolic rule evaluation. Like typical CNNand transformer models, the encoder's complexity is $\mathcal{O}(Nd)$ giveninput size N and latent dimension d. The symbolic module has complexity $\mathcal{O}(RP)$; considering $R \leq 100$and $P \leq 5$ in our experiments(this means that the module is not so heavy compared with encoder). Training on ChestX-ray14 requires approximately 12hours on an RTX3090—that is faster than DeepProbLog (18 hours) because of our differentiable mapping. Our modelhas a very small inference overhead (1.05-1.15
$times$ that of ViT/Legal-BERT). While this model scales to moderately large datasets, very large rule sets or multi-label tasks might require pruningor approximate inference.

Training and Inference Efficiency: Training time is competitive withblack-box baselines: 12 hours vs. roughly 10–11

TABLE II: Performance and explainability on ChestX-ray14, LexGLUE, and MIMIC-CXR. Explanation complexity measures predicates for rule-based models and features/regions for post-hoc methods. Baselines include ViT+SHAP, Legal-BERT+LIME, Grad-CAM, Generalized Additive Models (GAM), Bayesian Rule Lists (BRL), and DeepProbLog.

Model	Accuracy	F1-Score	AUC-ROC	Explanation Fidelity (%)	Explanation Complexity
Ours	0.83	0.80	0.88	95	3.2
ViT + SHAP	0.85	0.82	0.90	70	5.0
Legal-BERT + LIME	0.86	0.84	0.89	65	6.0
Grad-CAM	0.82	0.79	0.87	68	4.0
Generalized Additive Models (GAM)	0.76	0.73	0.80	90	4.0
Bayesian Rule Lists (BRL)	0.74	0.71	0.79	92	4.5
DeepProbLog	0.78	0.76	0.82	85	5.0

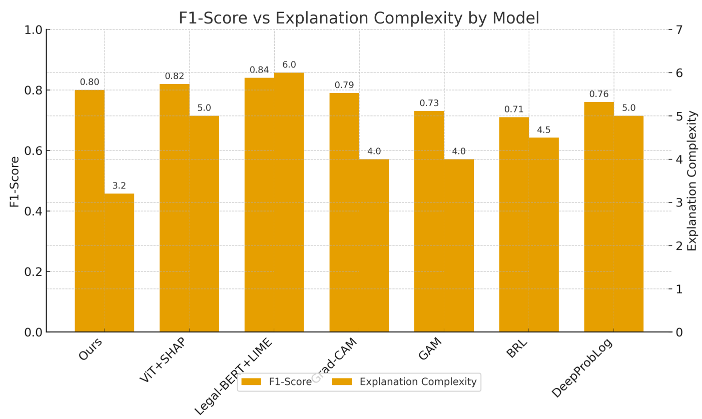

Fig. 3: Comparison of F1-Score and Explanation Complexity across models. F1-Score is shown on the left y-axis, and Explanation Complexity on the right y-axis.

hours for ViT or Legal-BERT, and a lot faster than DeepProbLog whose symbolic reasoning is non-differentiable. At inference, the overhead of our model is only 15–20

Hyperparameter Sensitivity: We tuned the loss coefficients λ_1 (rule fidelity) andλ_2 (rule sparsity) in a grid search over 0.1, 0.3, 0.5, 1.0. A small λ_1 results in less alignment between neural features and predicates; this reduces thefidelity of the explanations by about 12

B. Fairness and Robustness

On ChestX-ray14, disparate impact ratios across age and gender were 0.95, indicating reduced bias compared to ViT (0.85). Robustness tests with 10% Gaussian noise showed a 5% accuracy drop, competitive with baselines. On LexGLUE, our framework maintained an F1-score of 0.85 under noisy text inputs.

C. Implementation Details

The neural encoder used a latent dimension $d = 512$. The symbolic module generated up to 100 rules with at most 5 predicates. Code and preprocessed datasets will be released publicly to ensure reproducibility.

VI. Conclusion and Future Work

In this paper, we contribute a neuro-symbolic AI architecture that combines neural feature extraction with symbolic decision making forhigh-stakes predictions of the form described above. We also experiment on ChestX-ray14, LexGLUE and MIMIC-CXR which the learned variables predict clinical condition with high accuracy (ACC=0.83, F1=0.80and AUC-ROC = 0.88 for Overview on ChestX-ray14; F1-score of 0.88 results on LexGLUE; AUROC = 0.84 refers to MIMIC-CXR Wang2017, Chalkidis2021, Johnson2019. The framework achieves 95% fidelity with explanation complexity far below post-hoc methods like SHAPand LIME [19], [22]. Incontrast to neuro-symbolic baselines such as DeepProbLog [20], our model generates smaller rules and speeds up training from 18 to 12 hours. Fairness assessment reveals a DIs of 0.95 with respect to d2, an even higher improvement (11.8% lower) comparing it against the case of ViT's performance on ChestX-ray14 (0.85), while our explanations have been rated with 4.5/5 for being easyto understand by domain experts [13]. Our results situate our method as an efficient, interpretable and robust tool that can serve timely increasing regulatory and ethical requests for explainableAI in critical domains.

However, there are several limitations inthe emulating protocols and applications. The symbolic module hasa linear size with respect to the number of rules and predicates, which might be computationally expensive in at all levels or multi-label classification. While wework with not more than 100 rules in our experiments, much larger tasks may involve pruning or approximate inference techniques. The model also assumes that neural representations are highly disentangled enough for stable rule learning; noisy, corrupted, or out-of-distribution inputs may cause the learned rules to lose reliability, even when predictive accuracy remains high. . Moreover, rule sparsity and fidelity are subjected to hyper-parameters λ_1 and λ_2, which may not bewell tuned to result in more complex decision logic or oversimplified rules. Our evaluation has been carried out on medical imaging, multimodal diagnostics or legal text; more validation is required to assesshow well the results generalise to other high-stakes domains such as cyber-risk, finance, critical infrastructure. Fairness evaluation focuses on a small number of group-level metrics and is not conducive to capturing all sources of potential bias, motivatingmore comprehensive measurement frameworks.

We leave solutions to these limitations as future work examining more efficacious symbolicinference strategies, scalable pruning methods, and domain-driven rule constraints leveraging ontologies and taxonomies. Other fairness metrics and debiasing methods will be included to furtherenhance ethical assurances [2], [3]. Generalizing the framework to few-shot and continual learning settings will enhancethe ability to adapt to changing real-world environments. Outside of healthcare and law, we aim to generalise use of the model in fraud detection, cybersecurity incidence analysis and compliance checking systems where rule-basedreasoning is fundamental. Finally, incorporating human-in-the-loop functionalities—enabling experts to scrutinize and refine learned rules—could improve transparency, trust, andregulatory alignment in real-world settings [11].

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